Section One: Title, objective, responsibilities

Title or working title of project (in the form of a question, commission or design brief)

To design and develop a Python program that can generate a random maze, solve it using a pathfinding algorithm and then train an AI agent to find the optimal path (shortest route), with visualisation to demonstrate the learning process. This AI maze solver is for my younger brother Aleks who is participating in a maze solving competition that requires a maze memorisation solution.

Project objectives (eg, what is the question you want to answer? What do you want to learn how to do? What do you want to find out?):

By the end of this project I want to have an AI that can autonomously solve a randomly generated maze. To achieve this, I want to first deepen my understanding of reinforcement learning, pathfinding algorithms and maze generation techniques. I also want to explore how AI maze solving compares to mathematics based maze solving

To meet these aims, I set myself these objectives:

- **Develop a maze generation system that can produce random mazes of varying sizes and complexities using a combination of algorithms present in discrete mathematics.**
- **Implement Dijkstra's algorithm to solve the maze deterministically to provide a benchmark to compare that AI against and evaluate its performance.**
- **Create and train an AI model using reinforcement learning (q-function/deep q-function) that learns to navigate the maze using trial and error.**
- **Observe any maze solving techniques some of the later generations of AI have used to solve the maze and compare them with maze theory used nowadays (game theory).**

If it is a group project, what will your responsibilities be?

- **Not a group project.**

Section Two: Reasons for choosing this project

Reasons for choosing the project (eg, links to other subjects you are studying, personal interest, future plans, knowledge/skills you want to improve, why the topic is important):

I am currently studying maths, further maths, physics and computer science (this project is related to my studies.) I also have a strong personal interest in programming and computer science and I make small programming projects in my spare time. Therefore, undertaking a large, independent programming project like this was attractive to me. I am also applying to Oxbridge and an EPQ project of such technical depth could help demonstrate my academic curiosity and my ability to apply theory into real problems.

Also AI is currently one of the most rapidly advancing fields in recent technology and is helping to transform a variety of industries. It is not only important to know how to use AI, but it is also crucial that we have a rough understanding of how AI works. This will enable us to design better systems and help us to interact with current models more effectively.

More specifically, I will be using a reinforcement learning model to train my AI, a branch of AI that focuses on training models through trial and error. Reinforcement learning is one of the foundational training methods used in models like ChatGBT so my project is very relevant to modern day technology. Through this project, I aim to greatly expand my knowledge of artificial intelligence while also greatly improving my skills in programming and problem solving skills.

Section Three: Activities and timescales	

Activities to be carried out during the project (eg, research, development and analysis of ideas, writing, data collection, numerical analysis, rehearsal techniques, production meetings, production of final outcome, administration, evaluation, preparing for the presentation, etc):	How long this will take:
06/06/25: Complete a Draft Project Proposal In here I aim to clearly set out my aim of developing the python program as well as my reason for selecting the project. I will also clarify the timescales that I need to follow in order to successfully complete this project as well as the necessary research required.	4 hours
02/06/25 - 22/09/25: Update Activity Log Throughout my EPQ I will constantly update my activity log to have an accurate record of my development process. I will ensure that any entry into the activity log will not only be informative but also reflective on the particular entry - including what I learned or what challenges I overcame.	6-7 hours
02/06/25 - 22/09/25: Update Research Sources I will keep track of all my sources of research in a tabular format. In each research source, not only will I provide the research source, but I will also go to further depths to evaluate the reliability of the source as well as how useful it is to this project. This will be done by doing deeper research into the authors of a potential source using external sources.	5-6 hours
18/06/25: Consult at least one subject specialist In the session we went through the project. My teacher firstly verified that the project did not coincide with my computer science coursework. We also discussed the feasibility of my plan, primarily surrounding the AI reinforcement learning component and whether to do a deep-q-learning model or a standard tabular Q-Learning model. He also verified that the project was of A Level standard.	30 minutes
20/06/25: Write a draft Introduction to my project In my draft introduction I plan to lay out the premise of the project as well as some of the motivations for doing this project - for example my personal desire to help my brother out with his robotics competition. I also plan to lay out some rough development objectives to ensure that the flow of my EPQ makes sense to the majority of readers.	2-3 hours
02/07/25: Write my Research Review Before actually writing my research review I plan to get a firm understanding of all the theory and potential algorithms that I could be using in this project. This will involve reading articles, watching videos and going to teachers of computer science and further maths to consolidate my knowledge. Then, once I roughly understand what I will be doing, I will explain it in a manner that is easy to read and understand for most users. I will also begin to make links between the knowledge I have acquired and its application in my final artifact.	14-15 hours
15/07/25: Set up the basic project I study computer science in Peter Symonds College and the language they use there is C#. I have also not used Python in quite a while (2-3 years) so I feel it will be necessary to make a couple quick low scale programs just to recap the python syntax. I believe this will save a lot of time in the future. After I am finished, I will set up the folder and create the program. Inside the program, I	2 hours

will design and create a few 2d arrays containing the first graphs to make testing the algorithms easier. **15/07/25: Finish coding Breadth First Traversal** This algorithm will be responsible for verifying that any randomly generated graph is connected. Alongside just coding the algorithm, I need to test it on the graphs that I have created in order to verify that it actually works. At this stage I also plan to set up some classes in order to ensure that my program is well structured and follows the standard object oriented programming paradigm.	2-3 hours
17/07/25: Finish Prim's Algorithm This algorithm will be responsible for providing the maze-like structure to the graph. This will be done by converting the graph into a minimum spanning tree. Similarly to the Breadth First Traversal, I will be testing it on the graphs that I have created to verify that this algorithm works. After this, I will implement some additional functions to re-add a few of the arcs removed in order to further improve the maze-like structure (mazes usually have loops.)	2-3 hours
17/07/25: Finish Dijkstra's Algorithm to find the shortest path At this stage, I plan to create Dijkstra's algorithm to solve the generated maze. This will probably involve creating 2 separate functions, one to perform the forward pass and one to perform the backward pass. This solution will act as a benchmark for evaluating the performance of my AI later. Also, at this stage, I should be able to generate a random maze of any size and so I should be able to use these mazes to test the performance of the function. The solution found by my algorithm can be checked against the solution found by an online Dijkstra's algorithm tool.	4-5 hours
23/07/25: Create the AI model to solve the maze I plan to design and implement a reinforcement learning model (either Q-learning or Deep Q-learning) that learns to solve the maze through trial and error. This AI model will belong in its own separate class to further ensure that I am still upholding this object oriented programming paradigm. Before I begin the UI, it is crucial that I compare the efficiency of the AI against Dijkstra's algorithm.	8 hours
07/08/25: User interface complete Once the core functionality is developed, I can begin to focus on the visualisation. The libraries Matplotlib and NetworkX allow me to display the maze, perform Dijkstra's and carry out AI training and testing in real time. This whole stage is to allow the user to visually see and understand how the algorithms work. **There will be a big gap here primarily due to other commitments. I need to continue revising for the TMUA. I also need to work on my computer science coursework as it has similar deadlines to the EPQ. Alongside this, I am abroad for a couple weeks. I will try to stay informed on tech over this time so I don't forget the core concepts behind my code however, this will be difficult.**	5-6 hours
01/10/25 - 14/10/25: School PC's security issues Try to get the program working on the school PC's. I think this will be a difficult challenge because the school PC's have a host of safety and security measures that prevent me from running this software. I don't wish to breach any safety measures, but rather, convert any program that I make into a form that is considered safe by the PC's. Amidst this time I will be revising for the TMUA a lot so apart from this, I doubt I will do much else around the EPQ. **15/10/25 - 26/10/25: Detailed documentation process**	3-4 hours

Although the timing seems counterintuitive as you'd expect the documentation to go alongside coding, I plan to do the documentation of the program after I have finished coding. I believe it is crucial that I do this because I believe that a context switch (from coding to documentation) could be detrimental to the efficiency of my coding. Doing the project in this style allows me to immediately lead on from one algorithm to the next ensuring that I always understand my project while programming. In the documentation process, I will provide a detailed breakdown of each function. I will provide the approximate timings in **blue**: **15/10/25:** By this date I will finish the write up for my breadth first traversal code. **17/10/25:** Finish the write up and explanation of the maze generation process (Prim's.) **19/10/25:** Write up my implementation of Dijkstra's algorithm. **21/10/25 - 23/10/25:** I plan to have fully described how my code uses reinforcement learning to train the AI by the end of this period. **25/10/25:** Write up my usage of external libraries to provide the visualisation for my AI Maze solver. **15/11/25: Conclusion** This will be a short piece of work summarising my EPQ and giving a reader a satisfying ending to my project. I have given myself a buffer window of about 20 days between finishing the development report and writing the conclusion because I am pretty certain there will be a variety of little things (such as activity log or research sources) that I will need to catch up on. **20/11/25: Evaluation** This will be a much more detailed piece of work than the conclusion as it will involve a much deeper level of reflection on my whole project. I don't often write evaluations (for any of my A-Levels) so I will try to structure my evaluation in a manner similar to an exemplar A* EPQ artefact. **25/11/25: Abstract** I will write a brief summary at the start of my project to introduce any readers to my project. Counterintuitively, this should be done at the very end of the project so that your summary is most accurate (you know exactly what you have done.) I plan to get near finishing my project before December because I can potentially have interviews over December.	8-9 hours 1 hour 3-4 hours 30 minutes

Section Four: Resources

What resources will you need for your research, write up and presentation (eg, libraries, books, journals, equipment, rehearsal space, technology and equipment, venue, physical resources, finance):

College lesson notes and presentations
I will use my computer science and mathematics lesson materials (presentations/ documents/videos) to further reinforce my understanding of graph theory and algorithms. Because these materials are sourced from the college, I am certain that they are of A-Level standard.

Public programming documentation
This primarily relates to the libraries I will be using in my program. Both NetworkX and Matplotlib have in depth documentation surrounding the program in their library. Using certain portions of the library as shown in the website will enable me to use their visualisation tools.

Youtube tutorials and public lectures
I personally find watching spoken lectures and tutorials effective in getting a broad idea of the content I will be learning. I think this will be especially useful when learning about the Q-Function reinforcement learning models as I have no prior experience in this area of computer science.

Websites and academic articles
I will use websites after I have a rough idea of what I am learning about to get into fine details of the subject. I believe this will be particularly useful when describing algorithms such as Prim's or Dijkstra's and will enable me to use more advanced, better-fitting terminology.

What your areas of research will cover?

Discrete mathematics
This is the study of mathematical structures that are considered to be discrete. More specifically, I will be focused on graph theory, which involves concepts like nodes, arcs, trees and networks and this will form the foundation for my project. The algorithms that I will be using in this project are also based entirely on graph theory.

Python programming
This will be the programming language I will use in this project. I will use it to create algorithms, structure my code using OOP and integrate libraries which will create my final artefact.

AI (Reinforcement learning)
This is a branch of AI that involves training models through trial and error - using rewards and penalties. The core focus of my project is to use reinforcement learning to solve a maze.

Data Visualisation and human interaction
This area is all about focusing on how I present the algorithms and the AI visually so that it is easily digestible to the uninformed user. By researching the best Python UI libraries I can make sure that my project is as engaging as possible.

Ethics and potential implications of AI

This area is all about focusing on the long term implications of AI. This may be from a safety perspective, where AI could be used to automate heavy machinery, which could potentially become a safety hazard to humans. Ethically, AI raises concerns around mass surveillance and privacy, as well as potential job cuts due to increased automation.